

Inequality Proofs — Solutions

Warm-up

Question 1a:

To prove $a^2 + b^2 > 2ab$ for distinct real numbers a and b , we start with the property that the square of any real number is non-negative.

$$\begin{aligned}(a - b)^2 &\geq 0 \\ a^2 - 2ab + b^2 &\geq 0 \\ a^2 + b^2 &\geq 2ab\end{aligned}$$

If $a \neq b$, then $(a - b)^2 > 0$, which implies $a^2 + b^2 > 2ab$.

Question 1b:

We want to prove $x^2 + 4y^2 > 4xy$. We can treat $4y^2$ as $(2y)^2$.

$$\begin{aligned}(x - 2y)^2 &\geq 0 \\ x^2 - 2(x)(2y) + (2y)^2 &\geq 0 \\ x^2 - 4xy + 4y^2 &\geq 0 \\ x^2 + 4y^2 &\geq 4xy\end{aligned}$$

Assuming $x \neq 2y$, the inequality is strict: $x^2 + 4y^2 > 4xy$.

Question 1c:

To prove the Arithmetic Mean-Geometric Mean (AM-GM) inequality for two variables, $\frac{a+b}{2} \geq \sqrt{ab}$, we assume $a, b \geq 0$.

$$\begin{aligned}(\sqrt{a} - \sqrt{b})^2 &\geq 0 \\ (\sqrt{a})^2 - 2\sqrt{a}\sqrt{b} + (\sqrt{b})^2 &\geq 0 \\ a - 2\sqrt{ab} + b &\geq 0 \\ a + b &\geq 2\sqrt{ab} \\ \frac{a+b}{2} &\geq \sqrt{ab}\end{aligned}$$

Question 2a:

The triangle inequality $|a + b| \leq |a| + |b|$ states that the absolute value of the sum of two real numbers is always less than or equal to the sum of their absolute values. In simpler terms, it means that the shortest distance between two points is a straight line; taking a detour through a third point (represented by the addition of b) cannot result in a shorter path than going directly.

Question 2b:

In geometry, the triangle inequality theorem states that for any triangle, the sum of the lengths of any two sides must be greater than the length of the remaining side. If we represent the sides as vectors $\vec{a}$ and $\vec{b}$, the third side is $\vec{a} + \vec{b}$. The length of the third side $|\vec{a} + \vec{b}|$ must be less than or equal to the sum of the lengths of the other two sides $|\vec{a}| + |\vec{b}|$. (Diagram description: A triangle with vertices A, B, C . Side AB has length $|a|$, side BC has length $|b|$, and side AC has length

$|a + b|$. The inequality $|a + b| \leq |a| + |b|$ represents the fact that the path $A \rightarrow C$ is shorter than or equal to the path $A \rightarrow B \rightarrow C$.

Question 3a:

Using the triangle inequality $|a + b| \leq |a| + |b|$, let $b = 1$.

$$\begin{aligned} |a + 1| &\leq |a| + |1| \\ |a + 1| &\leq |a| + 1 \end{aligned}$$

Question 3b:

We apply the triangle inequality to the expression $|2x + 3y|$.

$$\begin{aligned} |2x + 3y| &\leq |2x| + |3y| \\ |2x + 3y| &\leq |2| \cdot |x| + |3| \cdot |y| \\ |2x + 3y| &\leq 2|x| + 3|y| \end{aligned}$$

Question 3c:

We apply the triangle inequality to the expression $|3m - 4n|$. Note that $|-4n| = 4|n|$.

$$\begin{aligned} |3m - 4n| &\leq |3m| + |-4n| \\ |3m - 4n| &\leq |3| \cdot |m| + |-4| \cdot |n| \\ |3m - 4n| &\leq 3|m| + 4|n| \end{aligned}$$

Skill-building

Question 1a:

To prove $a + \frac{1}{a} \geq 2$ for $a > 0$, we start with the known inequality $(a - 1)^2 \geq 0$.

$$\begin{aligned} (a - 1)^2 &\geq 0 \\ a^2 - 2a + 1 &\geq 0 \\ a^2 + 1 &\geq 2a \end{aligned}$$

Dividing both sides by a (since $a > 0$):

$$\begin{aligned} \frac{a^2 + 1}{a} &\geq \frac{2a}{a} \\ \frac{a^2}{a} + \frac{1}{a} &\geq 2 \\ a + \frac{1}{a} &\geq 2 \end{aligned}$$

Question 1b:

We use the identity $1 = \sin^2 x + \cos^2 x$ and the fact that $(\sin x - \cos x)^2 \geq 0$.

$$\begin{aligned}
(\sin x - \cos x)^2 &\geq 0 \\
\sin^2 x - 2 \sin x \cos x + \cos^2 x &\geq 0 \\
(\sin^2 x + \cos^2 x) - 2 \sin x \cos x &\geq 0 \\
1 - 2 \sin x \cos x &\geq 0 \\
1 &\geq 2 \sin x \cos x
\end{aligned}$$

Question 1c:

We use the identity $1 = \sin^2 x + \cos^2 x$ and the fact that $(\sin x + \cos x)^2 \geq 0$.

$$\begin{aligned}
(\sin x + \cos x)^2 &\geq 0 \\
\sin^2 x + 2 \sin x \cos x + \cos^2 x &\geq 0 \\
(\sin^2 x + \cos^2 x) + 2 \sin x \cos x &\geq 0 \\
1 + 2 \sin x \cos x &\geq 0 \\
2 \sin x \cos x &\geq -1 \\
-1 &\leq 2 \sin x \cos x
\end{aligned}$$

Question 1d:

We start with the fact that $(e^x - 1)^2 \geq 0$ for all real x .

$$\begin{aligned}
(e^x - 1)^2 &\geq 0 \\
(e^x)^2 - 2(e^x)(1) + 1^2 &\geq 0 \\
e^{2x} - 2e^x + 1 &\geq 0 \\
e^{2x} + 1 &\geq 2e^x
\end{aligned}$$

Question 2:

To prove $|a + b| \leq |a| + |b|$, we square both sides. Since both sides are non-negative, the inequality is preserved.

$$\begin{aligned}
|a + b|^2 &\leq (|a| + |b|)^2 \\
(a + b)^2 &\leq |a|^2 + 2|a||b| + |b|^2 \\
a^2 + 2ab + b^2 &\leq |a|^2 + 2|ab| + |b|^2
\end{aligned}$$

Since $a^2 = |a|^2$ and $b^2 = |b|^2$, we can subtract these from both sides:

$$\begin{aligned}
2ab &\leq 2|ab| \\
ab &\leq |ab|
\end{aligned}$$

Since any number is always less than or equal to its absolute value, $ab \leq |ab|$ is always true. Thus, $|a + b| \leq |a| + |b|$ is proven.

Question 3a:

We use the reverse triangle inequality $|x - y| \geq ||x| - |y||$. However, the question asks to prove $|x - y| \leq ||x| - |y||$. Let's check the validity. If $x = 2, y = 1$, then $|2 - 1| = 1$ and $||2| - |1|| = 1$. If $x = 2, y = -1$, then $|2 - (-1)| = 3$ and $||2| - |-1|| = 1$. In this case, $3 \leq 1$ is false. The

question likely intended the reverse triangle inequality $|x - y| \geq ||x| - |y||$ or $|x - y| \leq |x| + |y|$. Assuming the intended proof was $|x - y| \leq |x| + |y|$:

$$\begin{aligned} |x - y| &= |x + (-y)| \\ &\leq |x| + |-y| \\ &\leq |x| + |y| \end{aligned}$$

Question 3b:

We apply the triangle inequality repeatedly.

$$\begin{aligned} |a + b + c| &= |(a + b) + c| \\ &\leq |a + b| + |c| \\ &\leq (|a| + |b|) + |c| \\ &\leq |a| + |b| + |c| \end{aligned}$$

Question 3c:

We use the property $|\ln(xy)| = |\ln x + \ln y|$.

$$\begin{aligned} |\ln(x \sin x)| &= |\ln x + \ln(\sin x)| \\ &\leq |\ln x| + |\ln(\sin x)| \end{aligned}$$

This follows directly from the triangle inequality $|a + b| \leq |a| + |b|$ where $a = \ln x$ and $b = \ln(\sin x)$.

Question 4a:

We are given $-1 \leq \sin(1/x) \leq 1$. We multiply the entire inequality by x^2 . Since $x^2 \geq 0$ for all x , the inequality signs remain the same.

$$-x^2 \leq x^2 \sin(1/x) \leq x^2$$

As $x \rightarrow 0$, we evaluate the limits of the bounding functions:

$$\begin{aligned} \lim_{x \rightarrow 0} (-x^2) &= 0 \\ \lim_{x \rightarrow 0} (x^2) &= 0 \end{aligned}$$

Since the function $x^2 \sin(1/x)$ is squeezed between two functions that both approach 0, by the Squeeze Theorem, $\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0$.

Question 4b:

We know that $-1 \leq \sin x \leq 1$ for all x .

$$\frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x} \quad (\text{for } x > 0)$$

As $x \rightarrow \infty$, we evaluate the limits of the bounding functions:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{-1}{x} &= 0 \\ \lim_{x \rightarrow \infty} \frac{1}{x} &= 0 \end{aligned}$$

Since the function $\frac{\sin x}{x}$ is squeezed between two functions that both approach 0, by the Squeeze Theorem, $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$.

Question 4c:

We know that $-1 \leq \sin x \leq 1$ for all x .

$$\text{Since } e^{-x} > 0, \quad -e^{-x} \leq e^{-x} \sin x \leq e^{-x}$$

As $x \rightarrow \infty$, we evaluate the limits of the bounding functions:

$$\begin{aligned} \lim_{x \rightarrow \infty} (-e^{-x}) &= 0 \\ \lim_{x \rightarrow \infty} (e^{-x}) &= 0 \end{aligned}$$

Since the function $e^{-x} \sin x$ is squeezed between two functions that both approach 0, by the Squeeze Theorem, $\lim_{x \rightarrow \infty} e^{-x} \sin x = 0$.

Easier Exam Questions

Question 1:

Given $a, b < 0$ and $a > b$.

$$(A) \ a - b > 0 \implies \frac{1}{a - b} > 0 \quad (\text{Since } a > b)$$

Thus, (A) is false. Let's check the others.

$$(B) \ \frac{a}{b} - \frac{b}{a} = \frac{a^2 - b^2}{ab} = \frac{(a - b)(a + b)}{ab}$$

Since $a > b, a - b > 0$. Since $a, b < 0, a + b < 0$ and $ab > 0$.

$$\implies \frac{(+)(-)}{(+)} < 0$$

(B) is true. Checking (C):

$$(C) \ a + b > 2b \implies a > b \quad (\text{True by given})$$

Checking (D):

$$(D) \ 2a > 3b \implies a > 1.5b. \quad \text{Since } a > b \text{ and } b < 0, 3b < 2b < a. \quad \text{True.}$$

Therefore, (A) is the statement that might be false (it is actually always false given the conditions).

Question 2i:

We start with the property that the square of any real number is non-negative.

$$\begin{aligned} (a - b)^2 &\geq 0 \\ a^2 - 2ab + b^2 &\geq 0 \\ a^2 + b^2 &\geq 2ab \end{aligned}$$

Question 2ii:

We want to show $\frac{x}{x^2+4} \leq \frac{1}{4}$ for $x \geq 0$.

$$\begin{aligned}\frac{x}{x^2+4} &\leq \frac{1}{4} \\ 4x &\leq x^2+4 \\ 0 &\leq x^2-4x+4 \\ 0 &\leq (x-2)^2\end{aligned}$$

Since $(x-2)^2 \geq 0$ is always true for $x \geq 0$, the original inequality $\frac{x}{x^2+4} \leq \frac{1}{4}$ is also true.

Question 3i:

For positive real numbers a, b , we use the AM-GM inequality.

$$\frac{a+b}{2} \geq \sqrt{ab}$$

This is a standard result for positive real numbers.

Question 3ii:

We want to prove $(a+b+c+d)^2 \geq 4(ac+bc+bd+ad)$.

$$\begin{aligned}\text{LHS} &= (a+b)^2 + 2(a+b)(c+d) + (c+d)^2 \\ &= (a+b)^2 + 2(ac+ad+bc+bd) + (c+d)^2\end{aligned}$$

Since $(a+b)^2 \geq 0$ and $(c+d)^2 \geq 0$, we have:

$$\text{LHS} \geq 2(ac+ad+bc+bd)$$

To get to $4(ac+ad+bc+bd)$, we need to use the fact that $a, b, c, d > 0$.

$$(a+b+c+d)^2 = (a+b)^2 + 2(a+b)(c+d) + (c+d)^2$$

Using the inequality $x^2 + y^2 \geq 2xy$ where $x = a+b$ and $y = c+d$:

$$(a+b)^2 + (c+d)^2 \geq 2(a+b)(c+d)$$

Substituting this back into the expansion:

$$\begin{aligned}(a+b+c+d)^2 &\geq 2(a+b)(c+d) + 2(a+b)(c+d) \\ &= 4(a+b)(c+d) \\ &= 4(ac+ad+bc+bd)\end{aligned}$$

Question 3iii:

We want to prove $(a+b+c+d)^2 \geq \frac{8}{3}(ab+ad+bc+cd+bd+ac)$.

Let $S = a+b+c+d$. Then $S^2 = a^2 + b^2 + c^2 + d^2 + 2(ab+ac+ad+bc+bd+cd)$.

We know $a^2 + b^2 + c^2 + d^2 \geq \frac{1}{3}((a+b+c+d)^2)$ is not quite right.

Using the inequality $3(a^2 + b^2 + c^2 + d^2) \geq (a + b + c + d)^2$ is not helpful. Let's use $a^2 + b^2 + c^2 + d^2 \geq 2(ab + ac + ad + bc + bd + cd)/3$ is not standard. 's use the identity $(a + b + c + d)^2 = a^2 + b^2 + c^2 + d^2 + 2 \sum_{sym} ab$.

We know $a^2 + b^2 + c^2 + d^2 \geq \frac{2}{3} \sum_{sym} ab$ is not generally true.

Actually, let's use the property that for n variables, $n \sum x_i^2 \geq (\sum x_i)^2$.

$$\begin{aligned} \text{LHS} &= (a + b + c + d)^2 \\ \text{RHS} &= \frac{8}{3} \sum_{sym} ab \end{aligned}$$

Consider the difference $S^2 - \frac{8}{3} \sum ab = a^2 + b^2 + c^2 + d^2 + 2 \sum ab - \frac{8}{3} \sum ab = a^2 + b^2 + c^2 + d^2 - \frac{2}{3} \sum ab$.

We know $3(a^2 + b^2 + c^2 + d^2) \geq 2 \sum_{sym} ab$ is not the standard form.

Let's use the fact that $a^2 + b^2 + c^2 + d^2 \geq$

$\frac{2}{3} \sum_{sym} ab$ is true for positive real numbers. , let's use the inequality $a^2 + b^2 + c^2 + d^2 \geq \frac{2}{3} \sum_{i < j} x_i x_j$ for $n = 4$.

$$\begin{aligned} \text{LHS} &= a^2 + b^2 + c^2 + d^2 + 2 \sum ab \\ \text{Since } a^2 + b^2 + c^2 + d^2 &\geq \frac{2}{3} \sum ab, \quad \text{LHS} \geq \frac{2}{3} \sum ab + 2 \sum ab = \frac{8}{3} \sum ab \end{aligned}$$

Question 4i:

We are given $\frac{a+b}{2} \geq \sqrt{ab}$. For positive real numbers a, b, c , we have:

$$\begin{aligned} \frac{a+b}{2} &\geq \sqrt{ab} \implies a+b \geq 2\sqrt{ab} \\ \frac{b+c}{2} &\geq \sqrt{bc} \implies b+c \geq 2\sqrt{bc} \\ \frac{c+a}{2} &\geq \sqrt{ca} \implies c+a \geq 2\sqrt{ca} \end{aligned}$$

Multiplying these three inequalities (since all terms are positive):

$$\begin{aligned} (a+b)(b+c)(c+a) &\geq (2\sqrt{ab})(2\sqrt{bc})(2\sqrt{ca}) \\ (a+b)(b+c)(c+a) &\geq 8\sqrt{a^2b^2c^2} \\ (a+b)(b+c)(c+a) &\geq 8abc \end{aligned}$$

Question 4ii:

Let x, y, z be the sides of a triangle. Then $x + y > z$, $y + z > x$, and $z + x > y$. Let $a = y + z - x$, $b = z + x - y$, and $c = x + y - z$. Since x, y, z are sides of a triangle, $a, b, c > 0$. From part (i), we know $(a+b)(b+c)(c+a) \geq 8abc$.

Substitute a, b, c :

$$a + b = (y + z - x) + (z + x - y) = 2z$$

$$b + c = (z + x - y) + (x + y - z) = 2x$$

$$c + a = (x + y - z) + (y + z - x) = 2y$$

Substituting these into the inequality from (i):

$$(2z)(2x)(2y) \geq 8(y + z - x)(z + x - y)(x + y - z)$$

$$8xyz \geq 8(y + z - x)(z + x - y)(x + y - z)$$

$$xyz \geq (y + z - x)(z + x - y)(x + y - z)$$

Harder Exam Questions

Question 1i:

We want to prove $\frac{x}{\sqrt{x-1}} \geq 2$ for $x > 1$.

Let $f(x) = \frac{x}{\sqrt{x-1}}$. We want to show $f(x) - 2 \geq 0$.

$$\frac{x}{\sqrt{x-1}} - 2 \geq 0$$

$$\frac{x - 2\sqrt{x-1}}{\sqrt{x-1}} \geq 0$$

Since $\sqrt{x-1} > 0$ for $x > 1$, we only need to prove $x - 2\sqrt{x-1} \geq 0$.

Let $u = \sqrt{x-1}$. Then $u^2 = x - 1 \implies x = u^2 + 1$.

Substitute into the numerator:

$$(u^2 + 1) - 2u \geq 0$$

$$(u - 1)^2 \geq 0$$

Since $(u - 1)^2 \geq 0$ is always true, the original inequality $\frac{x}{\sqrt{x-1}} \geq 2$ is proven.

Question 1ii:

From part (i), we know that for any $t > 1$, $\frac{t}{\sqrt{t-1}} \geq 2$. Squaring both sides:

$$\frac{t^2}{t-1} \geq 4$$

Now, consider the expression $\frac{a^2}{b-1} + \frac{b^2}{a-1}$. We can use the Titu's Lemma (a form of Cauchy-Schwarz):

$$\sum \frac{x_i^2}{y_i} \geq \frac{(\sum x_i)^2}{\sum y_i}.$$

$$\begin{aligned} \frac{a^2}{b-1} + \frac{b^2}{a-1} &\geq \frac{(a+b)^2}{(b-1) + (a-1)} \\ &= \frac{(a+b)^2}{a+b-2} \end{aligned}$$

Let $S = a + b$. We want to show $\frac{S^2}{S-2} \geq 8$.

$$\begin{aligned}\frac{S^2}{S-2} - 8 &\geq 0 \\ \frac{S^2 - 8(S-2)}{S-2} &\geq 0 \\ \frac{S^2 - 8S + 16}{S-2} &\geq 0 \\ \frac{(S-4)^2}{S-2} &\geq 0\end{aligned}$$

Since $a > 1$ and $b > 1$, $S = a + b > 2$, so $S - 2 > 0$. Since $(S - 4)^2 \geq 0$, the inequality holds. Thus, $\frac{a^2}{b-1} + \frac{b^2}{a-1} \geq 8$.

Question 2i:

We want to prove $a^2 + b^2 + c^2 \geq ab + ac + bc$ for positive real numbers a, b, c .

Start with the known inequalities:

$$\begin{aligned}a^2 + b^2 &\geq 2ab \\ b^2 + c^2 &\geq 2bc \\ c^2 + a^2 &\geq 2ca\end{aligned}$$

Summing these three inequalities:

$$\begin{aligned}(a^2 + b^2) + (b^2 + c^2) + (c^2 + a^2) &\geq 2ab + 2bc + 2ca \\ 2(a^2 + b^2 + c^2) &\geq 2(ab + bc + ca) \\ a^2 + b^2 + c^2 &\geq ab + bc + ca\end{aligned}$$

Question 2ii:

We want to prove $a^3 + b^3 + c^3 \geq 3abc$. This is a classic result of the AM-GM inequality for three variables.

$$\begin{aligned}\frac{a^3 + b^3 + c^3}{3} &\geq \sqrt[3]{a^3 b^3 c^3} \\ \frac{a^3 + b^3 + c^3}{3} &\geq abc \\ a^3 + b^3 + c^3 &\geq 3abc\end{aligned}$$

Question 2iii:

We want to prove $ac(a + c) + ab(a + b) + bc(b + c) \geq 6abc$.

Expand the expression:

$$a^2c + ac^2 + a^2b + ab^2 + b^2c + bc^2 \geq 6abc$$

Group the terms into pairs and apply AM-GM to each pair:

$$a^2b + ab^2 \geq 2\sqrt{a^2b \cdot ab^2} = 2\sqrt{a^3b^3} \quad (\text{Not quite})$$

Actually, apply AM-GM to each term:

$$a^2b + b^2c + c^2a \geq 3\sqrt[3]{a^2b \cdot b^2c \cdot c^2a} = 3\sqrt[3]{a^3b^3c^3} = 3abc$$

$$ab^2 + bc^2 + ca^2 \geq 3\sqrt[3]{ab^2 \cdot bc^2 \cdot ca^2} = 3\sqrt[3]{a^3b^3c^3} = 3abc$$

Summing these two results:

$$(a^2b + b^2c + c^2a) + (ab^2 + bc^2 + ca^2) \geq 3abc + 3abc$$

$$ac(a + c) + ab(a + b) + bc(b + c) \geq 6abc$$

Question 3i:

For distinct positive real numbers a, b , we have $(a - b)^2 > 0$.

$$(a - b)^2 > 0$$

$$a^2 - 2ab + b^2 > 0$$

$$a^2 + b^2 > 2ab$$

Question 3ii:

For distinct positive real numbers a, b, c , we apply the result from (i) to each pair:

$$a^2 + b^2 > 2ab$$

$$b^2 + c^2 > 2bc$$

$$c^2 + a^2 > 2ca$$

Summing these inequalities:

$$2(a^2 + b^2 + c^2) > 2(ab + bc + ca)$$

$$a^2 + b^2 + c^2 > ab + bc + ca$$

Question 3iii:

We want to prove $\frac{a^2b^2 + b^2c^2 + c^2a^2}{a+b+c} > abc$.

Multiply both sides by $(a + b + c)$:

$$a^2b^2 + b^2c^2 + c^2a^2 > abc(a + b + c)$$

$$a^2b^2 + b^2c^2 + c^2a^2 > a^2bc + ab^2c + abc^2$$

Let $x = ab, y = bc, z = ca$. The inequality becomes:

$$x^2 + y^2 + z^2 > xy + yz + zx$$

From part (ii), we know that for distinct a, b, c , the products ab, bc, ca are also distinct. Thus, $x^2 + y^2 + z^2 > xy + yz + zx$ is true. Therefore, the original inequality holds.

Question 4i:

For $a, b > 0$, we use the AM-GM inequality:

$$\begin{aligned}\frac{a}{b} + \frac{b}{a} &\geq 2\sqrt{\frac{a}{b} \cdot \frac{b}{a}} \\ \frac{a}{b} + \frac{b}{a} &\geq 2\sqrt{1} \\ \frac{a}{b} + \frac{b}{a} &\geq 2\end{aligned}$$

Question 4ii:

We use the result from (i) with a_i and 1. For each i , we have:

$$\begin{aligned}\frac{a_i}{1} + \frac{1}{a_i} &\geq 2 \implies a_i + 1 \geq 2\sqrt{a_i \cdot 1} \quad (\text{by AM-GM}) \\ 1 + a_i &\geq 2\sqrt{a_i}\end{aligned}$$

Multiplying these inequalities for $i = 1, 2, \dots, n$:

$$\begin{aligned}(1 + a_1)(1 + a_2) \dots (1 + a_n) &\geq 2\sqrt{a_1} \cdot 2\sqrt{a_2} \dots 2\sqrt{a_n} \\ (1 + a_1)(1 + a_2) \dots (1 + a_n) &\geq 2^n \sqrt{a_1 a_2 \dots a_n}\end{aligned}$$

Since $a_1 a_2 \dots a_n = 1$, we have $\sqrt{a_1 a_2 \dots a_n} = 1$.

$$(1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n$$

Question 4iii:

We use the AM-GM inequality on each term of the LHS:

$$\begin{aligned}\frac{a^2}{b} + b &\geq 2\sqrt{\frac{a^2}{b} \cdot b} = 2a \\ \frac{b^2}{c} + c &\geq 2\sqrt{\frac{b^2}{c} \cdot c} = 2b \\ \frac{c^2}{d} + d &\geq 2\sqrt{\frac{c^2}{d} \cdot d} = 2c \\ \frac{d^2}{a} + a &\geq 2\sqrt{\frac{d^2}{a} \cdot a} = 2d\end{aligned}$$

Summing these four inequalities:

$$\begin{aligned}\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a} + (a + b + c + d) &\geq 2a + 2b + 2c + 2d \\ \frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a} &\geq a + b + c + d\end{aligned}$$

Question 5i:

For $a, b, c > 0$, we use the result from the sum of squares inequality:

$$\begin{aligned}a^2 + b^2 &\geq 2ab \\ b^2 + c^2 &\geq 2bc \\ c^2 + a^2 &\geq 2ca\end{aligned}$$

Summing these:

$$\begin{aligned}2(a^2 + b^2 + c^2) &\geq 2(ab + bc + ca) \\ a^2 + b^2 + c^2 &\geq ab + bc + ca\end{aligned}$$

Question 5ii:

We want to prove $2(a^3 + b^3 + c^3) \geq bc(b + c) + ca(c + a) + ab(a + b)$.

$$\begin{aligned}\text{LHS} &= 2a^3 + 2b^3 + 2c^3 \\ \text{RHS} &= b^2c + bc^2 + c^2a + ca^2 + a^2b + ab^2\end{aligned}$$

We can group the terms as follows:

$$\begin{aligned}a^3 + b^3 &\geq ab(a + b) \quad (\text{Since } (a + b)(a - b)^2 \geq 0) \\ b^3 + c^3 &\geq bc(b + c) \\ c^3 + a^3 &\geq ca(c + a)\end{aligned}$$

Summing these three inequalities:

$$\begin{aligned}(a^3 + b^3) + (b^3 + c^3) + (c^3 + a^3) &\geq ab(a + b) + bc(b + c) + ca(c + a) \\ 2(a^3 + b^3 + c^3) &\geq bc(b + c) + ca(c + a) + ab(a + b)\end{aligned}$$